

Subject Name: Bioprocess Technology Laboratory

Subject No: BT69204

Prerequisite: Bioprocess Technology (BT30201) or Aspects of Biochemical Engineering (BT61201)

Objective: To impart practical hands-on training on the kinetics of enzyme catalyzed reactions, enzyme immobilization and performance evaluation of immobilization method and immobilized enzyme reactor

L-T-P 0-0-3

Credit 2

Content:

1. Determination of total and specific enzyme activity of Amylase (3 hours)
2. Determination of the kinetic constants K_m and V_{max} of amylase catalyzed starch hydrolysis under specified conditions (3 hours)
3. Determination of the kinetic constants, K_m and V_{max} , of starch hydrolysis catalyzed by immobilized (entrapped in calcium alginate beads) amylase under specified conditions. (6 hours)
4. Determination of the kinetic constants K_m and V_{max} of amylase immobilized in celite matrix (adsorption mechanism) (6 hours)
5. Immobilization of α -amylase in gelatin matrix by cross linking mechanism and determination of the kinetic constants K_m and V_{max} of immobilized enzyme (6 hours)
6. Performance of immobilized amylase enzymes in Batch reactor (3 hours)
7. Performance of immobilized amylase enzymes in CSTR operated with different flow rates (3 hours)
8. Performance of an immobilized enzyme packed bed reactor (PBR) operated at different flow rates for starch hydrolysis (6 hours)

Text books:

1. Biochemical Engineering Fundamentals, by James E. Bailey and David F. Ollis, McGraw-Hill International Editions.
2. Enzymes: A Practical Introduction to Structure, Mechanism, and Data Analysis, by Robert A. Copeland, John Wiley and Sons Inc.